

Complete Biology Curriculum

Comprehensive Academic Syllabus for Ordinary Level (O'Level) & Advanced Level (A'Level)

Institution: Ejikeme Masterclass

Platform: ejikeme-masterclass.github.io

Target: IGCSE / WAEC / Cambridge A-Level

PART 1: O'LEVEL BIOLOGY SYLLABUS

Theme 1: Cell Biology & Cellular Organization

1.1 Cell Structure, Function & Organization

- Plant vs. animal cell structures (organelles: nucleus, cell membrane, cytoplasm, mitochondria, chloroplasts, cell wall, vacuoles).
- Microscopy fundamentals: Light microscopy techniques vs. electron microscopy resolution capabilities.
- Specialized cell adaptations: Structure-function relationships in root hair cells, xylem vessels, red blood cells, and motor neurons.
- Hierarchical levels of biological organization: Cells, tissues, organs, organ systems, and whole organisms.

1.2 Movement of Substances into & out of Cells

- Diffusion: Concentration gradients, surface area-to-volume ratio, temperature influence, and distance factors.
- Osmosis: Water potential dynamics, solute concentration effects, turgor pressure in plants, plasmolysis, and crenation.
- Active Transport: Energy requirement (ATP), carrier protein uptake against concentration gradients (e.g., ion uptake in roots).

Theme 2: Biological Molecules & Enzymes

2.1 Biomolecules & Chemical Tests

- Chemical constituent elements of carbohydrates, lipids, and proteins (C, H, O, N, S, P).
- Synthesis and basic structural units: Glucose to starch/glycogen/cellulose; amino acids to proteins; glycerol + fatty acids to lipids.
- Qualitative biochemical testing procedures: Benedict's test (reducing sugars), Iodine solution (starch), Biuret test (proteins), and Ethanol emulsion test (fats/lipids).

2.2 Enzyme Actions & Kinetics

- Enzymes as biological catalysts: Activation energy reduction and specificity.
- Mechanism of action: The Lock and Key hypothesis, active site complementary structure, and substrate complex formation.
- Factors affecting catalytic rate: Temperature coefficient (Q_{10}), optimum pH levels, thermal denaturation, and kinetic energy trends.

Theme 3: Plant Physiology & Energetics

3.1 Plant Nutrition & Photosynthesis

- Photosynthesis equation: Balanced chemical reaction ($6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$) and energy conversion.
- Leaf micro-anatomy: Cuticle, upper/lower epidermis, palisade mesophyll, spongy mesophyll, stomatal complex, and vascular bundles.
- Limiting factors of photosynthetic rate: Light intensity, carbon dioxide concentration, ambient temperature, and chlorophyll availability.

3.2 Transport & Transpiration in Plants

- Xylem vs. Phloem structural features and functional differentiation (water/minerals vs. sucrose/amino acids translocation).
- Transpiration stream: Evaporation mechanisms, root pressure, xylem cohesion-tension, wilting, and environmental factors affecting transpiration rate.

Theme 4: Human Physiology & Homeostasis

4.1 Human Digestive System & Nutrition

- Balanced diet parameters, energetic demands, and nutritional deficiency conditions (scurvy, rickets, kwashiorkor).
- Alimentary canal organs, mechanical breakdown (mastication, peristalsis), and chemical digestion by specific hydrolases (amylases, proteases, lipases).
- Intestinal absorption: Microvilli structural adaptations in the ileum, nutrient absorption into capillaries, and lacteal transport.

4.2 Respiration & Respiratory Exchange

- Aerobic vs. Anaerobic respiration pathways: Energy yield comparisons and chemical pathways in yeast and human muscle tissue.
- Human respiratory apparatus: Trachea, bronchi, bronchioles, alveoli architecture, and intercostal/diaphragm ventilation mechanics.
- Gas exchange mechanics across alveolar membranes; biological consequences of tobacco smoke inhalation.

4.3 Human Circulation & Transport

- Internal heart anatomy, double circulatory pathways (systemic vs. pulmonary loops), and cardiac cycle dynamics.
- Structural comparison of blood vessel types: Arteries, arterioles, capillaries, venules, and veins.
- Blood tissue components: Red blood cells, white blood cells (phagocytes, lymphocytes), platelets, and plasma composition.

4.4 Excretion, Coordination & Homeostasis

- Excretory system: Kidney nephron structure, ultrafiltration, selective reabsorption, and urine formation.
- Homeostatic control principles: Negative feedback loops, thermoregulation mechanisms in human skin, and blood glucose control (insulin and glucagon actions).
- Nervous integration: Reflex arc, synaptic transmission, central vs. peripheral systems, and internal eye mechanics (accommodation, light response).

Theme 5: Genetics, Reproduction & Ecology

5.1 Reproduction & Development

- Asexual vs. sexual reproduction: Genetic variation outcomes, advantages, and limitations.
- Plant sexual reproduction: Flower structures, insect/wind pollination adaptations, double fertilization, and seed dispersal mechanisms.
- Human reproduction: Male/female anatomy, menstrual cycle hormonal regulation (FSH, LH, estrogen, progesterone), fertilisation, and placenta function.

5.2 Genetics, Inheritance & Natural Selection

- Chromosomes, DNA structure basics, genes, alleles, dominant/recessive traits, homozygous/heterozygous genotypes.
- Monohybrid cross prediction using Punnett squares, pedigree charts, codominance, and sex determination (XX/XY).
- Variation forms (continuous vs. discontinuous), gene/chromosomal mutations, natural selection principles, and antibiotic resistance development.

5.3 Ecology & Environmental Impact

- Ecosystem dynamics: Trophic levels, food chains, complex webs, and pyramids of numbers, biomass, and energy efficiency.
- Nutrient cycling pathways: The Carbon and Nitrogen cycles (nitrogen fixation, nitrification, denitrification, decomposition).
- Anthropogenic ecosystem impacts: Deforestation, greenhouse gas elevation, eutrophication, bioaccumulation, and biodiversity conservation.

PART 2: ADVANCED LEVEL (A'LEVEL) SYLLABUS

Theme 1: Advanced Biochemistry & Cell Biology

1.1 Molecular Biochemistry & Structural Biology Advanced

- Condensation and hydrolysis reactions in polymer formation.
- Carbohydrates: Monosaccharide stereoisomers (α - and β -glucose), disaccharides (maltose, sucrose, lactose), structural glycosidic linkages (1,4- and 1,6-) in starch, glycogen, and cellulose.
- Lipid chemistry: Triglyceride esterification, saturated vs. unsaturated fatty acid chains, amphipathic phospholipids, and membrane bilayer dynamics.
- Protein architecture: Primary sequence, secondary structures (α -helix, β -pleated sheet), tertiary folding (hydrogen, ionic, disulfide bonds, hydrophobic interactions), and quaternary protein complexes (e.g., hemoglobin, collagen).
- Water physics: Dipolar nature, extensive hydrogen bonding, high specific heat capacity, latent heat of vaporization, and solvent properties.

1.2 Ultrastructure & The Cell Cycle Advanced

- Eukaryotic organelle biochemistry: Endoplasmic reticulum (smooth and rough), Golgi apparatus, lysosomes, peroxisomes, chloroplasts, and mitochondria.
- Prokaryotic cell organisation: Peptidoglycan cell walls, 70S ribosomes, nucleoid DNA, plasmids, and flagellar motors.
- Advanced microscopy: Transmission (TEM) vs. Scanning (SEM) electron microscopy, resolution limits, eyepiece graticule calibration via stage micrometers.
- Cell fractionation and differential centrifugation protocols for organelle isolation.
- Cell division dynamics: Mitosis stages, interphase (G_1 , S , G_2) checkpoints, cyclin-dependent kinase control, oncogenes, and tumorigenesis.

Theme 2: Enzyme Kinetics & Bioenergetics

2.1 Advanced Enzyme Kinetics Advanced

- Induced-fit theory of enzyme catalysis vs. static lock-and-key models.
- Michaelis-Menten kinetics: Determination and significance of maximum velocity (V_{max}) and Michaelis constant (K_m).
- Inhibitor dynamics: Competitive vs. non-competitive inhibition effects on V_{max} and K_m values, allosteric regulation, and end-product feedback loops.

2.2 Cellular Respiration & Metabolic Pathways Advanced

- Glycolysis: Phosphorylation, lysis, oxidation, and ATP yield in cytosol.
- The Link Reaction & Krebs Cycle: Mitochondrial matrix decarboxylation, dehydrogenation, production of reduced **NAD** and **FAD**, substrate-level phosphorylation.
- Oxidative Phosphorylation: Electron transport chain (ETC) complexes, chemiosmotic proton gradients across inner mitochondrial membrane, and ATP synthase mechanics.
- Anaerobic metabolic pathways: Lactate pathway vs. alcoholic fermentation; substrate energy yields and Respiratory Quotients (**RQ**).

2.3 Photosynthetic Photochemistry Advanced

- Light-dependent reaction: Photosystems I and II excitation, non-cyclic and cyclic photophosphorylation, photolysis of water ($2H_2O \rightarrow 4H^+ + 4e^- + O_2$), and NADP reduction.
- Light-independent reaction (Calvin Cycle): RuBisCO carbon fixation, glycerate-3-phosphate (GP) reduction to triose phosphate (TP), RuBP regeneration.
- Photorespiration dynamics and rate-limiting photosynthetic constraints under varied environmental regimes.

Theme 3: Molecular Genetics, Gene Expression & Biotechnology

3.1 Molecular Genetics & Protein Synthesis Advanced

- DNA replication mechanism: Semi-conservative replication, DNA helicase, DNA polymerase proofreading, replication fork directionality ($5' \rightarrow 3'$), and Okazaki fragments.
- Transcription & Post-Transcriptional Processing: RNA polymerase action, promoter regions, pre-mRNA splicing (introns vs. exons), spliceosomes, 5' capping, poly-A tail addition.
- Translation: Transfer RNA (tRNA) structure/aminoacyl-tRNA synthetase, ribosomal codon-anticodon pairing (**AUG** start, stop codons), polypeptide elongation.
- Gene expression regulation: Prokaryotic regulation (the *lac* operon system), eukaryotic transcription factors, epigenetics (histone acetylation, DNA methylation).

3.2 Inheritance Patterns, Population Genetics & Evolution Advanced

- Advanced Mendelian Genetics: Dihybrid inheritance ratios, autosomal linkage, recombination frequencies, epistasis (recessive/dominant), and Chi-squared (χ^2) statistical testing.
- Meiosis mechanism: Synapsis, homologous chromosome bivalent formation, crossing over (chiasmata formation), independent assortment, and non-disjunction errors.
- Population Genetics: The Hardy-Weinberg Principle ($p^2 + 2pq + q^2 = 1$), allele frequency shifts, genetic drift (founder effect, bottlenecks), selection types (stabilizing, directional, disruptive).
- Speciation pathways: Allopatric vs. sympatric isolation mechanisms and reproductive isolation barriers.

3.3 Recombinant DNA Technology & Applied Biotechnology Advanced

- Gene Cloning Techniques: Restriction endonucleases, reverse transcriptase synthesis, vector insertion (plasmids), DNA ligase, marker genes, host transformation.
- Polymerase Chain Reaction (PCR): Thermocycling stages (denaturation, annealing, extension via Taq polymerase).
- Analytical separation: Agarose gel electrophoresis for DNA fragments, Southern blotting, DNA sequencing protocols, DNA profiling/fingerprinting.
- Gene editing innovations: Principles of CRISPR-Cas9 endonuclease systems and genetic modification applications in medicine and agriculture.

Theme 4: Neurobiology, Hormonal Control & Homeostasis

4.1 Action Potentials, Synapses & Muscle Physiology Advanced

- Resting Membrane Potential: Na^+/K^+ ATPase pump activity, ion channel permeability, electrochemical equilibrium (-70mV).
- Action Potential dynamics: Voltage-gated Na^+/K^+ channel kinetics, depolarization, repolarization, hyperpolarization, absolute/relative refractory periods, and saltatory conduction along myelinated axons.
- Synaptic Transmission: Cholinergic synapse event cascade, calcium ion influx, exocytosis of acetylcholine, acetylcholinesterase breakdown, and summation (temporal vs. spatial).
- Muscular System: Microscopic structure of skeletal muscle myofibrils (sarcomeres, actin, myosin), sliding filament theory, troponin-tropomyosin complex, and cross-bridge cycles.

4.2 Advanced Homeostasis & Signal Transduction Advanced

- Renal Physiology: Ultrafiltration across glomerular capillaries (podocytes, basement membrane), selective reabsorption in proximal convoluted tubule, countercurrent multiplier mechanism in the Loop of Henle, ADH regulation via aquaporins in collecting ducts.
- Intracellular Signaling: Second messenger systems (adrenalin/glucagon binding, G-protein activation, adenylate cyclase, cyclic AMP production, protein kinase A cascades).
- Plant Signaling: Phytohormone action (auxins, gibberellins, abscisic acid), cell elongation mechanics via acid-growth hypothesis, and phototropism/geotropism pathways.

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